Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH. (AUTOMOBILE ENGINEERING)

END SEMESTER EXAMINATIONS, NOV 2019

SUBJECT: ERGONOMICS IN AUTOMOTIVE DESIGN [AAE 4026]

REVISED CREDIT SYSTEM
(27/11/2019)

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitable assumed.

1A. Define anthropometry and biomechanics with suitable examples. **(02)**

1B. Explain three major consideration in human capability and characteristics in the ergonomics consideration. **(03)**

1C. List and explain the NINE ergonomic engineer's responsibilities. **(05)**

2A. What is CTD? What causes CTD? **(02)**

2B. Explain the complete understanding of L5/S1 phenomenon in vehicle seat design. What is the effect of L5/S1 on armrest? **(03)**

2C. Explain the reason for the standard dimension / size, to be maintained, for the following along with SAE number: i) Seat cushion length ii) Seat width iii) Seatback height iv) Lateral location of the seat v) Bolster height. **(05)**

3A. Explain the following phrase with example: i) "What people don't understand, does not exist." ii) "Hands-free may not be risk-free." **(02)**

3B. List the difference between HPM and HPD. **(03)**

3C. Explain the following design considerations terms w.r.t developing the occupant package: i) Comfortable seated posture ii) Visibility of interior and exterior area iii) Operating controls iv) Entry and Egress space. **(05)**

4A. Explain the procedure for maximum hand reach as per the SAE J287 standard. **(02)**

4B. What are the advantages and disadvantages of touch screen display used in the vehicles? **(03)**

4C. What Is Field of View? Explain the following characteristics on which driver in a vehicle depends on: (i) the driver (ii) the vehicle (iii) the targets (iv) the environment. **(05)**

5A. List and explain problems during entry and exit. **(02)**

5B. With a neat sketch explain the running board design. **(03)**

5C. List and explain important vehicle dimensions during entry and exit. **(05)**